

Proof Buddy — API Reference (Testing Edition)

Complete HTTP API reference for writing contract tests, integration tests, and manual API checks. Documents **as-implemented** behavior from Django views and `urls.py`.

Status: De-facto / exploratory. No OpenAPI spec. Lock envelope **keys** in tests; do not lock exact error **strings** until team decides.

Related documentation

Document	Role
API_REFERENCE.md	This file — HTTP API reference: routes, params, and responses
DATA_FLOW.md	What happens inside Django after each API call (engine, cache, DB)
ARCHITECTURE.md	System overview: frontend, backend apps, auth, deployment
DEVELOPER_GUIDE.md	Proof engine internals, where to change rules, how to extend the system
TESTING_GUIDE.md	How to run backend and frontend tests
COURSE_APIS.md	Assignments/courses narrative spec with sample JSON (some paths differ from code; prefer this file for routes)
induction_api/README.md	Induction endpoint walkthrough with curl examples
4_API_reference.md	Legacy — minimal auth-only list; outdated paths (<code>register</code> , <code>/users/profile</code>); kept for old install-doc links

Table of contents

1. [Global conventions](#)
2. [Auth](#) — `/api/v1/auth/`
3. [Equational](#) — `/api/v1/equational/`
4. [Induction](#) — `/api/v1/induction/`
5. [Legacy proof](#) — `/api/v1/proof/`
6. [Proofs](#) — `/api/v1/proofs`
7. [Assignments](#) — `/api/v1/assignments/`
8. [Known issues](#)

Global conventions

Topic	Detail
Base URL	<code>/api/v1/</code>
Format	JSON request/response
Auth header	<code>Authorization: Token <accessToken></code>

Topic	Detail
Token source	<code>POST /api/v1/auth/signin</code> → <code>{ accessToken, username }</code>
Session cache	Equational: <code>equational_obj_{username}</code> . Induction: <code>induction_obj_{username}</code> , <code>induction_proof_{username}</code> . Legacy: <code>proofs_{username}</code>
Call order	Initialize session before rule application
Soft fail	Many proof endpoints return 200 + <code>"isValid": false</code>
Persistence	<code>get-proof-lines</code> may return <code>{ "hasProof": false }</code> until proof is saved/loaded

Canonical vs legacy

Prefix	Status	Client service
<code>/api/v1/equational/</code>	Canonical	<code>equationalService.js</code>
<code>/api/v1/induction/</code>	Canonical	<code>inductionService.js</code>
<code>/api/v1/proof/</code>	Legacy	<code>erService.js</code>
<code>/api/v1/proofs</code>	Sparse (2 routes)	<code>proofsService.js</code>
<code>/api/v1/auth/</code>	Canonical	<code>authService.js</code> , <code>userService.js</code>
<code>/api/v1/assignments/</code>	Canonical	<code>courseServices.js</code>

Auth

Prefix: `/api/v1/auth/` · **App:** `accounts/` · **Cache:** none

`POST /api/v1/auth/signup`

Field	Detail
Auth	No
Body	<code>email</code> (string), <code>username</code> (string), <code>password</code> (string), <code>is_instructor</code> (bool, optional)
Success 201	<code>{ "message": "Account created!" }</code>
Fail 400	<code>{ "message": { "<field>": ["..."] } }</code>

`POST /api/v1/auth/signin`

Field	Detail
Auth	No
Body	<code>username</code> (string), <code>password</code> (string)
Success 200	<code>{ "accessToken": string, "username": string }</code>
Fail 404	Unknown username: <code>{ "message": "Invalid username/password" }</code>

Field	Detail
Fail 400	Wrong password: { "message": "Invalid username/password" }

GET /api/v1/auth/profile

Field	Detail
Auth	Yes
Success 200	{ "username", "email", "is_student" } — is_student = not is_instructor
Fail 404	{ "message": "User not found" }

POST /api/v1/auth/logout

Field	Detail
Auth	Yes (token to invalidate)
Body	(none)
Success 200	{ "message": "Logged out" } — always 200 even if token delete fails

POST /api/v1/auth/activate-account

Field	Detail
Auth	No
Body	activation_key (string)
Success 200	{ "message": "Account activated" }
Fail 404	{ "message": "Invalid activation key" }

POST /api/v1/auth/forgot-password

Field	Detail
Auth	No
Body	email (string)
Success 200	{ "message": "Email sent" }
Fail 404	{ "message": "User not found" }

POST /api/v1/auth/reset-password

Field	Detail
Auth	No
Body	reset_key (string), password (string)
Success 200	{ "message": "Password reset" }

Field	Detail
Fail 404	{ "message": "Invalid activation key" }

POST /api/v1/auth/resend-activation-email

Field	Detail
Auth	No
Body	email (string)
Success 200	{ "message": "Email sent" }
Fail 404	{ "message": "User not found" }

Equational

Prefix: /api/v1/equational/ · **App:** equational_reasoning_api/ · **Cache key:** equational_obj_{username}

Recommended call order: save-proof or upload-proof or get-user-proof → set-current-proof → apply-rule/substitution → get-proof-lines → check-completion

POST /api/v1/equational/set-current-proof

Field	Detail
Auth	Yes
Body	lhsPremise (string, required), rhsPremise (string, required), definitions (array, default []), generics (array, default [])
Definition item	{ label, type, expression, is_default?, deletable? }
Generic item	{ label, type, restrictions }
Success 200	{ isValid: true, lhsPremise, rhsPremise, lhsJsonTree, rhsJsonTree }
Fail 400	{ isValid: false, errors: [...] }

Notes: Does not create DB proof. Preserves proof_id from cache if present. Hard-fail uses 400 (not soft 200).

POST /api/v1/equational/apply-rule

Field	Detail
Auth	Yes
Body	side (string, default "LHS"), currentRacket (string), rule (string), startPosition (int, default 0), selectedNode (int), substitution (string), lineNumber (int), supportRewriteComplexity (bool, default true)
Success 200	{ isValid, racket, jsonTree, rule, resultNodeId, errors }

Field	Detail
Soft-fail 200	<code>isValid: false, racket: "Error generating racket", jsonTree: {}</code>
Fail 400	<code>{ isValid: false, errors: [...] }</code>

Notes: Reloads lines from DB. Soft-fail on hidden/unimplemented definitions and misuse of `rewrite math/rewrite logic` in rule field.

POST /api/v1/equational/substitution

Same body/response as `apply-rule`. Maps `rule: "math" → "rewrite math", "logic" → "rewrite logic"`. Does not reload DB lines before acting.

DELETE /api/v1/equational/delete-line/<side>/<line_number>

Field	Detail
Auth	Yes
URL	<code>side (LHS/RHS), line_number (int)</code>
Success 200	<code>{ success: true, message: "Line cleared successfully" }</code>
Fail 400	<code>{ success: false, error: "..." }</code>

Notes: Only clears when `line_number > 0`.

POST /api/v1/equational/check-completion

Field	Detail
Auth	Yes
Body	<i>(empty)</i>
Success 200	<code>{ isComplete: bool, message: "Proof complete!" }</code>
Fail 400	<code>{ error: "..." }</code>

GET /api/v1/equational/get-proof-lines

Field	Detail
Auth	Yes
No proof_id 200	<code>{ hasProof: false }</code>
Unauthorized 403	<code>{ hasProof: false }</code>
Success 200	<code>{ hasProof: true, user, lhsAnchorGoal, rhsAnchorGoal, proofName, tag, proof_id, support_*, visible_rules, LHS[], RHS[], definitions[], generics[] }</code>

Line object: `lineNumber`, `racket`, `rule`, `jsonTree`, `selectedNode`, `resultNode`, `substitution`, `startPosition`, `hide_expression`, `hide_justification`, `errors`

POST /api/v1/equational/toggle-visibility

Field	Detail
Body	<code>side</code> (required), <code>lineNumber</code> (required), <code>field</code> (required: "expression")
Success 200	<code>{ success: true, line_number, side, field, new_value }</code>
Fail 400	<code>{ error: "..." }</code> — requires <code>proof_id</code> in cache

POST /api/v1/equational/toggle-visibility-premise

Same as `toggle-visibility` plus optional `setting_visibility` (bool) to set instead of toggle.

POST /api/v1/equational/validate-hidden-field

Field	Detail
Body	<code>side</code> , <code>lineNumber</code> (required); <code>studentRule</code> , <code>studentExpression</code> , <code>studentSelectedNode</code> (optional)
Success 200	<code>{ isValid, errors, hide_expression, hide_justification, message }</code>
Fail 400/404	<code>{ error: "..." }</code>

POST /api/v1/equational/validate-hidden-definition

Field	Detail
Body	<code>label</code> (required), <code>student_expression</code> (required)
Success 200	<code>{ isValid: true, expression }</code> or <code>{ isValid: false, message }</code>
Fail 400/500	<code>{ error: "..." }</code>

GET /api/v1/equational/proofs

Field	Detail
Query	<code>page</code> (default 1), <code>query</code> (name filter)
Success 200	<code>{ proofs[], totalPages, currentPage, hasNext, hasPrevious }</code>

Proof item: `id`, `name`, `tag`, `lhs`, `rhs`, `isComplete`, `proofLines[]`, `definitions[]`

POST /api/v1/equational/get-user-proof

Field	Detail
Body	<code>proof_id</code> (int, required)

Field	Detail
Success 200	<code>{ success: true, message: "Proof loaded and DB repaired" }</code>
Fail 400/404	<code>{ error: "..." }</code>

Notes: Loads into cache; call `get-proof-lines` after for data.

POST /api/v1/equational/clear-proof

Field	Detail
Body	<i>(empty)</i>
Success 200	<code>{ message: "Session cleared successfully" }</code>

Notes: Cache only; DB proof unchanged.

POST /api/v1/equational/discard-proof

Field	Detail
Success 200	<code>{ message: "Proof archived successfully" }</code>

Notes: Archives most recent active proof (`is_active=false`), clears cache.

POST /api/v1/equational/save-proof

Field	Detail
Body	<code>name</code> (required), <code>tag</code> , <code>lhsGoal</code> , <code>rhsGoal</code> (required), <code>leftPremise</code> , <code>rightPremise</code> , <code>leftRacketsAndRules[]</code> , <code>rightRacketsAndRules[]</code> , <code>definitions[]</code> , <code>generics[]</code>
Line object	<code>racket</code> , <code>rule</code> , <code>startPosition</code> , <code>substitution</code> , <code>selectedNode</code> , <code>resultNode</code> , <code>errors</code> , <code>jsonTree</code>
Success 201	<code>{ message: "Proof saved successfully", proofId }</code>
Fail 400	<code>{ message: "Error saving proof" }</code>

Notes: Reserved names rejected: `IH`, `length`, `append`, `reverse`. Does not update cache.

POST /api/v1/equational/delete-proof

Field	Detail
Body	<code>proof_id</code> (int, required)
Success 200	<code>{ success: true, message, cacheCleared }</code>
Fail 404	<code>{ error: "Proof not found" }</code>

PATCH /api/v1/equational/set-parameters

Field	Detail
Auth	Instructor only (proof must belong to user)
Body	<code>proof_id</code> (required); optional: <code>support_errors</code> , <code>support_current_lhs_rhs</code> , <code>support_ih</code> , <code>support_premise</code> , <code>support_rule_set</code> , <code>support_value_mapping</code> , <code>visible_rules</code> , <code>support_rewrite_complexity</code>
Success 200	All 8 param fields with current values
Fail 403/404	<code>{ error: "..."} </code>

GET /api/v1/equational/download-proof

Field	Detail
Query	<code>proof_id</code> (required)
Success 200	Export JSON: <code>proofType</code> : "equational", <code>name</code> , <code>tag</code> , <code>lhsGoal</code> , <code>rhsGoal</code> , <code>definition[]</code> , <code>support_*</code> , <code>lines.LHS[]</code> , <code>lines.RHS[]</code>

POST /api/v1/equational/upload-proof

Field	Detail
Body	Same shape as download response; <code>proofType</code> must be "equational"
Success 201	<code>{ proofId, proofName }</code>
Fail 400	<code>{ error: "..."} </code>

POST /api/v1/equational/save-comment

Field	Detail
Body	<code>side</code> , <code>line_number</code> , <code>role</code> ("student"
Success 200	<code>{ success: true }</code>

GET /api/v1/equational/get-comments

Field	Detail
Query	<code>side</code> , <code>line_number</code> (required)
Success 200	<code>{ student: "", instructor: "" }</code>

GET /api/v1/equational/get-comment-status

Field	Detail
Success 200	<code>{}</code> or <code>{ "LHS-0": true, "RHS-2": true, ... }</code>

Induction

Prefix: `/api/v1/induction/` · **App:** `induction_api/` · **Cache:** `induction_obj_{username}, induction_proof_{username}`

Recommended call order: `start-induction-proof` → `set-current-proof` → `apply-rule/substitution` → `check-completion`

Resume: `POST set-induction-session-by-id` with `{ proof_id }` → editing endpoints.

`POST /api/v1/induction/start-induction-proof`

Field	Detail
Body	<code>proof_name</code> (required), <code>induction_variable</code> (required), <code>anchor_value</code> (required), <code>leap_variable</code> (required); optional: <code>proof_tag</code> , <code>side</code> , <code>lhs_leap_goal</code> , <code>rhs_leap_goal</code> , <code>lhs_anchor_goal</code> , <code>rhs_anchor_goal</code> , <code>induction_type</code> ("integers")
Success 201	<code>{ message, proof_id, proof_name, proof_tag, generic_definition_created, generics_created, data, inductive_hypothesis_lhs, inductive_hypothesis_rhs }</code>
Fail 400/500	Validation or internal error

Notes: Archives existing active proof with same name. Call `set-current-proof` next.

`POST /api/v1/induction/clear-induction`

Field	Detail
Body	<i>(none)</i>
Success 200	<code>{ message, proof_name? }</code>

Notes: Soft-deletes active proof, clears `induction_obj_` cache.

`POST /api/v1/induction/new-proof`

Field	Detail
Auth	Yes
Success 200	<code>{ message: "Session cleared successfully" }</code>

Notes: Clears cache only; DB proof remains.

`POST /api/v1/induction/create-induction-proof/`

Field	Detail
Body	<code>induction_variable</code> , <code>anchor_value</code> , <code>leap_variable</code> , <code>lhs_expression</code> , <code>rhs_expression</code> (all required)
Success 201	Full <code>InductionProof</code> serializer object

Field	Detail
Fail 400	<code>{ error, details }</code>

Notes: Legacy simpler create; no `proof_name`.

GET `/api/v1/induction/get-induction-proofs/`

Field	Detail
Query	<code>query</code> (name filter, optional)
Success 200	<code>{ proofs: [InductionProof, ...] }</code>

GET `/api/v1/induction/proof/<proof_id>/`

Field	Detail
Success 200	Full <code>InductionProof</code> object
Fail 403	<code>{ hasProof: false }</code>
Fail 404	<code>{ error: "Proof not found" }</code>

POST `/api/v1/induction/set-current-proof`

Field	Detail
Body	<code>struct</code> (default "int"), <code>ivar</code> , <code>aval</code> , <code>lvar</code> , <code>lhsPremise</code> , <code>rhsPremise</code> (required); <code>definitions[]</code> , <code>generics[]</code>
Success 201	<code>{ isValid: true, errors: [], base: { LHS, RHS }, leap: { LHS, RHS } }</code> — each side has <code>racket</code> , <code>jsonTree</code>
Fail 400	<code>{ isValid: false, errors: [...] }</code>

POST `/api/v1/induction/apply-rule`

Field	Detail
Auth	Yes
Body	<code>case</code> (default "base" — also <code>anchor</code> , <code>leap</code> , etc.), <code>side</code> (default "LHS"), <code>currentRacket</code> , <code>rule</code> , <code>startPosition</code> , <code>selectedNode</code> , <code>substitution</code> , <code>lineNumber</code> , <code>supportRewriteComplexity</code>
Success 200	<code>{ isValid, racket, errors, jsonTree, lineNum, resultNodeId, rule }</code>
Fail 400	<code>{ isValid: false, errors: [...] }</code>

DELETE `/api/v1/induction/delete-line/<case>/<side>/<line_number>`

Field	Detail
Success 200	Empty body

Field	Detail
Fail 500	<code>{ error: "..."} </code>

Notes: Clears line (does not remove row). Only when `line_number > 0`.

POST /api/v1/induction/check-goal

Field	Detail
Body	<code>case</code> (default "base"), <code>side</code> (default "LHS"), <code>goal</code> (required)
Success 200	<code>{ isValid, errors, jsonTree }</code>

Notes: Uses fresh engine; does not persist. Used to validate goals before start.

POST /api/v1/induction/substitution

Same body pattern as `apply-rule`. Maps `math/logic` to rewrite rules.

POST /api/v1/induction/check-completion

Field	Detail
Body	<code>case</code> (default "base")
Success 200	<code>{ isComplete, label: "BASE CASE" }</code>

Notes: `overallComplete` requires both cases complete and no hidden fields (students).

GET /api/v1/induction/get-proof-lines

Field	Detail
Success 200	<code>{ base: { LHS[], RHS[] }, leap: { LHS[], RHS[] } }</code>

Line keys: `racket`, `rule`, `startPosition`, `selectedNode`, `resultNode`, `lineNumber`, `substitution`, `jsonTree`, `errors`, `hide_expression`, `hide_justification`

GET /api/v1/induction/get-current-proof

Field	Detail
No session 200	<code>{ hasProof: false }</code>
Success 200	<code>{ hasProof: true, inductionVariable, anchorValue, leapVariable, lhsAnchorGoal, rhsAnchorGoal, lhsLeapGoal, rhsLeapGoal, inductiveHypothesisLHS, inductiveHypothesisRHS, currentSide, isAnchorCase, proofName, tag, proof_id, support_* }</code>

DELETE /api/v1/induction/clear-all-proof-lines

Field	Detail
Success 200	{ message, deleted_count? }

POST /api/v1/induction/set-induction-session-by-id

Field	Detail
Auth	Yes
Body	proof_id (required)
Success 200	{ success: true }
Fail 403	{ hasProof: false }

POST /api/v1/induction/delete-proof

Field	Detail
Auth	Yes
Body	proof_id (required)
Success 200	{ success: true, message, cacheCleared }

GET /api/v1/induction/check-name-conflict

Field	Detail
Auth	Yes
Query	name (optional)
Success 200	{ conflict: bool, type: "Induction"

PATCH /api/v1/induction/set-parameters

Same shape as equational `set-parameters` (instructor only, `proof_id` required).

GET /api/v1/induction/download-proof

Field	Detail
Query	proof_id (required)
Success 200	Export with <code>proofType</code> : "induction", goals, IH fields, <code>lines.base</code> , <code>lines.leap</code>

POST /api/v1/induction/upload-proof

Field	Detail
Body	Download export shape; <code>proofType</code> must be "induction"
Success 201	{ proofId, proofName }

POST /api/v1/induction/validate-hidden-field

Field	Detail
Body	<code>side, case, lineNumber</code> (required); <code>studentExpression, studentRule, studentSelectedNode</code>
Success 200	<code>{ isValid, errors, hide_expression, hide_justification, message }</code>

POST /api/v1/induction/validate-hidden-definition

Field	Detail
Body	<code>label, student_expression</code> (required)
Success 200	<code>{ isValid, expression? }</code> or <code>{ isValid: false, message }</code>

POST /api/v1/induction/toggle-visibility

Field	Detail
Body	<code>side, lineNumber, field, case</code> (all required)
Success 200	<code>{ success, line_number, side, field, new_value }</code>

POST /api/v1/induction/toggle-visibility-premise

Same plus optional `setting_visibility` (bool).

Comment endpoints

Endpoint	Method	Params/body	Response
<code>save-comment</code>	POST	<code>side, line_number, role, comment</code>	<code>{ success: true, created? }</code>
<code>get-comments</code>	GET	query: <code>side, line_number</code>	<code>{ student, instructor }</code>
<code>get-comment-status</code>	GET	—	<code>{}</code> or <code>{ "LHS-0": true, ... }</code>

Legacy proof

Prefix: `/api/v1/proof/` · **App:** `racket_api/` · **Cache:** `proofs_{username}`

Modern equivalents exist under `/equational/`. Still used for definitions/generics via `erService.js`.

Method	Path	Body / params	Response keys	Notes
POST	<code>set-proof</code>	<code>definitions[], generics[], leftGoalChecked, lhsGoal, rightGoalChecked, rhsGoal, LHS[], RHS[]</code> (lines: <code>currentRacket, rule, startPosition</code>)	201: <code>{ isValid: "True", errors: [] }</code> — string "True"	Call first to init cache

Method	Path	Body / params	Response keys	Notes
POST	er-generate	side, currentRacket, rule, startPosition	{ isValid, racket, errors, jsonTree, lineNum }	Soft-fail 200
POST	er-substitution	+ substitution	Same as er-generate	
POST	check-goal	name, tag, side, goal, loadedProofId?	{ isValid, errors, jsonTree }	Soft-fail on name conflict
DELETE	delete-line/<side>	URL: side	Empty 200	
POST	er-clear	—	Empty 200	Clears cache
POST	er-save	See save body below	{ message } 201	Persists to DB
POST	er-complete	Same as er-save	Empty 200	Sets isComplete
GET	proofs	query: page, query	{ proofs, totalPages, currentPage, hasNext, hasPrevious }	
GET	proofs/<proof_id>	URL: proof_id	Full loaded proof object	Loads into cache
POST	er-definitions	label, type, expression, notes, expression_hidden?	Definition object 201	
GET	get-definitions	—	Array of definitions	
GET	use-definition/<label>	URL: label	{ message }	GET not POST
POST	edit-definition/	label, type, expression, notes, expression_hidden?	Updated definition	Trailing slash
DELETE	delete-definition/<label>/	URL: label	Empty 200	DB + cache
DELETE	remove-definition/<label>/	URL: label	Empty 200	Cache only
GET	get-generics	—	Array with enabled	
POST	create-generic	label, type, restrictions?, notes?	Generic object 201	
GET	use-generic/<id>	URL: id	Empty or { message }	

Method	Path	Body / params	Response keys	Notes
DELETE	<code>remove-generic/<id></code>	URL: <code>id</code>	Empty 200	Disable in session
DELETE	<code>delete-generic/<id></code>	URL: <code>id</code>	Empty 200	DB + session

Save/complete body (`er-save`, `er-complete`): `name`, `tag`, `lhsGoal`, `rHSGoal`, `leftPremise`, `rightPremise`, `leftRacketsAndRules[]`, `rightRacketsAndRules[]`, `definitions[]`, `generics` (dict keyed by label)

Proofs

Prefix: `/api/v1/proofs` (no trailing slash)

Method	Path	Body	Response	Notes
DELETE	<code>delete-proof/<id></code>	URL: <code>id</code>	{ <code>message</code> : "Proof successfully deleted" } 200	No auth check
PUT	<code>edit=proof/<id></code>	<code>name</code> , <code>tag</code> , <code>lhs</code> , <code>rhs</code> , <code>created_at</code> , <code>isComplete</code>	{ <code>message</code> : "Proof successfully updated" } 200	URL typo <code>edit=proof</code> . Client calls <code>edit-proof</code> . No auth

Assignments

Prefix: `/api/v1/assignments/` · **App:** `assignments/` · **Auth:** All endpoints require token (401 without)

Full narrative spec also in [COURSE_APIS.md](#) — **prefer paths below** where they differ.

Courses

Method	Path	Body / params	Roles	Success
GET	<code>courses</code>	—	Any	200: course array (instructor vs student shape differs)
POST	<code>courses</code>	<code>name</code> (req), <code>students[]</code> , <code>generate_join_code</code> , <code>expiration_date</code>	Instructor	201 + optional <code>join_code</code>
GET	<code>courses/<course_id></code>	URL: <code>course_id</code>	Owner, enrolled student, superuser	200: single course
PATCH	<code>courses/<course_id></code>	{ <code>action</code> : "regenerate_code" } OR partial: <code>name</code> , <code>is_active</code> , <code>term</code> , <code>description</code> , <code>join_code_expires_at</code>	Owner, superuser	200

Term validation: `^(Spring|Summer|Fall|Winter) \d{4}$`

403 messages: "You are not authorized to view/manage this course."

Assignments CRUD

Method	Path	Body	Roles	Success
GET	<code><course_id></code>	URL: course id	Owner, student, superuser	200: [AssignmentSerializer]
POST	<code>`` (root)</code>	<code>title,</code> <code>description,</code> <code>due_date,</code> <code>course,</code> <code>proofs[]</code>	Instructor (own course)	201
DELETE	<code>assignments/detail/<assignment_id></code>	URL: id	Owner, superuser	204
PATCH	<code>assignments/detail/<assignment_id></code>	Partial assignment fields + <code>proofs[]</code>	Owner, superuser	200

Proofs array item: `{ id, type: "equationalproof"|"inductionproof", name, order }`

AssignmentSerializer: `{ id, title, description, due_date, course, proofs[] }`

Student proof status: "Not Started" | "In Progress" | "Completed"

Progress & library

Method	Path	Roles	Response
GET	<code>assignments/<assignment_id>/progress</code>	Instructor, superuser	<code>{ columns[], students[] }</code> with per-proof statuses: "not started", "in progress", "complete", "late", "error"
GET	<code>instructor/library</code>	Any (own proofs)	<code>[{ id, name, type,</code> <code>displayType, tag }]</code>

Roster & enrollment

Method	Path	Body	Notes
POST	<code>check-user</code>	<code>{ student: string }</code>	username or email; 200 empty or 404
POST	<code>add-student</code>	<code>{ course: int, student: string }</code>	201 invitation; 409 disambiguation

Method	Path	Body	Notes
POST	<code>remove-student</code>	<code>{ course: int, student: string }</code>	204
POST	<code>join-course</code>	<code>{ code: string }</code>	200 + course object
POST	<code>leave-course</code>	<code>{ course: int }</code>	200

Invitations

Method	Path	Body	Response
GET	<code>courses/<course_id>/invitations</code>	—	<code>[CourseInvitationSerializer]</code>
DELETE	<code>courses/<course_id>/invitations</code>	<code>{ invitation_id }</code>	204
GET	<code>invitations/me</code>	—	Pending invitations for caller
POST	<code>invitations/me</code>	<code>{ invitation_id, action: "accept" }</code>	<code>"reject" }</code>

Sharing

Method	Path	Body / query	Notes
GET	<code>assignments/shares</code>	query: <code>course_id</code> (required)	<code>{ incoming[], sent[] }</code>
POST	<code>assignments/shares</code>	Send: <code>{ source_course_id, target_course_id, title, description?, due_date, proofs[] }</code> OR respond: <code>{ share_request_id, action }</code>	201 or 200
DELETE	<code>assignments/shares</code>	<code>{ share_request_id }</code>	Sender only
GET	<code>instructors/share-targets</code>	—	<code>[{ id, displayName, courses[] }]</code> — instructors only

Start assignment proof

Method	Path	Body	Response
POST	<code>assignments/<assignment_id>/start-assignment-proof</code>	<code>{ proof_id, proof_type: "equationalproof" }</code>	<code>"inductionproof" }</code>

Known issues

Issue	Detail
Three proof APIs	equational + induction (canonical) vs legacy <code>proof/</code>

Issue	Detail
edit-proof mismatch	Client: PUT .../edit-proof/<id>. Server: edit=proof/<id>
/proofs no auth	delete and edit have no ownership checks
Legacy set-proof	Success returns "isValid": "True" (string, not bool)
Legacy use-definition	GET method, not POST
COURSE_APIS.md drift	Some paths differ (e.g. share GET uses query param course_id)
Error strings	Not stable — assert structure only in contract tests
DRF permissions	Many endpoints lack explicit @IsAuthenticated; unauthenticated calls get broken behavior rather than clean 401

Source files

Prefix	Routes	Views
auth	accounts/urls.py	accounts/views.py
equational	equational_reasoning_api/urls.py	equational_reasoning_api/views.py
induction	induction_api/urls.py	induction_api/views.py
legacy proof	racket_api/urls.py	racket_api/views.py
proofs	proofs/urls.py	proofs/views.py
assignments	assignments/urls.py	assignments/views.py

Integration test references:

- django_server/equational_reasoning_api/test_integration.py
- django_server/induction_api/tests.py (+ test_*.py siblings)
- django_server/assignments/tests/